

Enhancing Multi-Agent Emotional Support Conversations for Code-Mixed Hinglish Help-Seekers with Cross-Lingual Experience Retrieval and Register-Aware Response Selection

Roshan Raj (12341830)
Indian Institute of Technology Bhilai
roshanr@iitbhilai.ac.in

Abstract—MultiAgentESC is a recent multi-agent framework for emotional support conversation (ESC) in which LLM agents analyse the help-seeker’s state, deliberate on support strategies with retrieved similar cases, and debate to select a response. It is designed, prompted and evaluated for English only, while many help-seekers in India express distress in code-mixed Hindi–English (Hinglish). We present CodeMixESC, which keeps the three-stage pipeline and adds four components without fine-tuning the LLM agents: a Code-Mix Profiler that measures the seeker’s register with word-level language identification and the Code-Mixing Index (CMI); a cross-lingual experience retriever, a multilingual sentence encoder fine-tuned contrastively on Hinglish–English pairs so that Hinglish queries retrieve the English case bank without translating it; register-aware generation and selection, which adds a language and cultural fit criterion to debate, voting and refinement; and a Register Gate that re-refines a response whose CMI departs from the seeker’s by more than a tuned threshold, adding at most one LLM call per turn. We also build ESConv-HiEn, a parallel Roman-script Hinglish version of the ESConv test set at two mixing levels that keeps the strategy labels, enabling a controlled robustness study. On ESConv-HiEn Heavy the fine-tuned retriever keeps 53% of the cases retrieved for the English original, against 27% for the base paper’s encoder, and CodeMixESC’s replies reach ROUGE-L 9.24 against 5.66 for MultiAgentESC with a smaller register gap and at most one extra LLM call per turn.

I. INTRODUCTION

Mental health care in India faces a severe shortage of professionals; the National Mental Health Survey 2015–16 reported a treatment gap of more than 60% for almost every mental disorder [1]. Emotional Support Conversation (ESC) systems [2] aim to reduce this gap with dialogue agents that comfort people using professional support strategies such as *Question*, *Reflection of feelings* and *Providing Suggestions*. MultiAgentESC [3] is a strong recent system in this line: specialised LLM agents analyse the user’s psychological state, deliberate on strategies using retrieved similar cases, and debate to select the best response.

People express emotions most authentically in their preferred language [4], and for many Indian users this is Hinglish typed in Roman script, e.g. “*yaar result aaya, bahut tension ho rahi hai, ghar pe kaise bataun*”. Recent benchmarks show that LLMs still struggle to reason about and generate code-

switched text [5]. For such users MultiAgentESC retrieves its experience with an English-only encoder, selects and refines responses without considering the user’s register, and has never been evaluated on code-mixed input. Code-mixing is a realistic distribution shift: handling it requires cross-lingual representation learning, robustness evaluation under a controlled shift, and generation towards a measurable target, the language mix.

Contributions. (i) CodeMixESC, which extends MultiAgentESC with a Code-Mix Profiler, a contrastively fine-tuned cross-lingual retriever, register-aware generation and selection, and a Register Gate (Section III); (ii) ESConv-HiEn, a parallel Hinglish version of the ESConv test set at two mixing levels with strategy labels (Section IV); (iii) a controlled evaluation of retrieval robustness, response quality, register match, strategy stability and efficiency against prompting baselines, MultiAgentESC and a translate-pivot pipeline, with ablations (Sections V–VI).

II. RELATED WORK

Emotional support conversation. ESConv [2] defines eight support strategies and provides strategy-annotated conversations. MultiAgentESC [3] organises LLM agents with AutoGen [6] into dialogue analysis, strategy deliberation with retrieved cases (Sentence-BERT [7]), and response generation with debate, reflection and voting. **Code-mixed NLP and mental health.** HingBERT [8] provides Hinglish language models and a word-level language identifier; PHINC [9] is a parallel Hinglish–English corpus. Srivastava et al. [4] introduce a Hinglish counselling dataset but only perform emotion recognition; Kumar and Chakraborty [10] generate code-mixed responses in general conversations without support strategies. To the best of our knowledge no prior work studies strategy-based emotional support generation for code-mixed help-seekers. **Cross-lingual sentence embeddings.** Multilingual Sentence-BERT models are obtained by knowledge distillation [11]; LaBSE [12] is trained on translation pairs. Contrastive training with in-batch negatives [13] is the standard way to adapt such encoders to a new pair of domains. **Register accommodation.** Communication Accommodation Theory [14] holds that converging towards a partner’s speech style

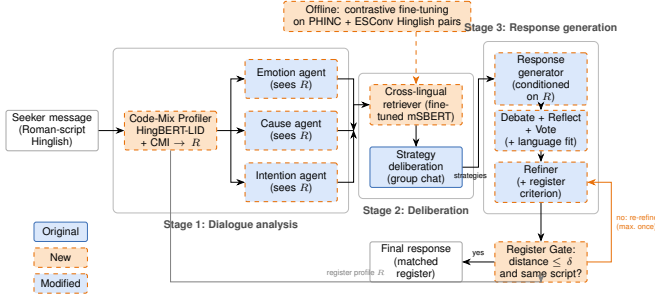


Fig. 1. CodeMixESC. Components in bold are new; components marked +R are modified to receive the seeker’s register profile R .

builds rapport, while diverging creates social distance, which motivates matching the seeker’s language mix.

III. METHOD

CodeMixESC keeps the three stages of MultiAgentESC (Fig. 1): (i) dialogue analysis, where a decision maker judges whether the dialogue reflects the user’s emotion, its cause and the user’s coping intention, and if so emotion, cause and intention agents analyse it; (ii) strategy deliberation, where three agents choose strategies in a round-robin group chat over the ten most similar ESConv cases; (iii) response generation, where one response per strategy is debated, reflected on and voted for, a judge breaks ties, and a refiner polishes the winner. Early turns and non-complex dialogues are answered by a single zero-shot agent. The decision maker, the deliberation prompt and the judge are unchanged.

A. Code-Mix Profiler

Every seeker utterance is tagged word by word with HingBERT-LID [8]; numbers, punctuation, emojis, URLs, mentions and hashtags are language independent, and Devanagari words are Hindi by script. The Code-Mixing Index [15] is

$$\text{CMI} = \begin{cases} 1 - \frac{\max_i w_i}{n - u}, & n > u \\ 0, & n = u \end{cases} \quad (1)$$

where w_i is the number of words of language i , n the number of tokens and u the number of language-independent tokens. Because a single short reply (“yes”) gives an unstable CMI, the profile is pooled over all seeker utterances so far (each utterance is tagged separately, and long texts are processed in chunks so that no word is truncated). The profiler outputs the register profile $R = (\text{CMI}_s, \text{dominant language, script})$, which every later agent receives together with an instruction to interpret common Hinglish distress expressions (*tension*, *ghabrahat*, *log kya kahenge*) in context rather than literally.

B. Cross-lingual experience retrieval

The base system encodes the seeker’s last message with the English encoder all-roberta-large-v1 and retrieves the ten most similar seeker posts of the English case bank (13,484 post–response pairs of the ESConv training conversations). We

replace it with paraphrase-multilingual-mpnet-base-v2 [11], fine-tuned on Hinglish–English pairs (h_i, e_i) with the Multiple Negatives Ranking loss [13]

$$\mathcal{L} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\cos(h_i, e_i)/\tau)}{\sum_{j=1}^B \exp(\cos(h_i, e_j)/\tau)}, \quad (2)$$

where B is the batch size and τ a temperature. Training pairs come from PHINC [9] and from ESConv training utterances rewritten into Hinglish by an LLM (Section IV). The English case bank is encoded once with the new model, so English experience is reused for Hinglish queries without translating the case bank.

C. Register-aware generation and selection

The response generator receives R and the instruction to reply “in the same script with a similar Hindi–English mix” while keeping the assigned strategy and the 30-word limit; the retrieved English examples are marked as illustrations of the strategy only. Debate, reflection (the voting round) and the refiner add a fourth criterion, *language and cultural fit*: “Does the response match the seeker’s language, script and level of Hindi–English mixing, and does it sound natural?”, so that a well-written but register-mismatched candidate can lose the vote. The single-agent path receives the same register instruction.

D. Register Gate

After refinement the gate profiles the response (Algorithm 1). If its register is more than δ away from the seeker’s or the script differs, the response is rewritten once with an explicit register instruction that states the measured mismatch (e.g. “about 5% Hindi words, while the user’s messages have about 62%”). The rewrite is kept only if its register is closer to the seeker’s, so the gate can never worsen the match, and it adds at most one LLM call per turn; δ is tuned on the development set. With two languages $\text{CMI} = \min(h, 1-h)$ for the share h of Hindi words, so the CMI gap cannot distinguish a mostly-Hindi reply from a mostly-English one (an 82%-Hindi reply to an English seeker has a CMI gap of only 0.18). The gate therefore measures $d = |h_r - h_s|$, which equals $|\text{CMI}_r - \text{CMI}_s|$ whenever both texts lean towards the same language and is larger exactly when the dominant language—part of R —flips. Because an English/Hindi homograph (*to*, *me*, *hi*) can be tagged Hindi in English text, a seeker is treated as code-mixing only with at least two unambiguous Hindi words (at least 5% of their words); otherwise the target register is plain English ($h_s = 0$).

E. Implementation

All agents are open-weight LLMs used only through prompting, as in the base paper: Gemma (gemma-4-26b-a4b-it) through the Gemini API free tier at temperature 0, with an OpenAI-compatible backend (e.g. Ollama) as an alternative. The AutoGen pipeline is re-implemented on a cached, rate-limited client with an exact replica of AutoGen’s round-robin group chat; all base prompts

Algorithm 1 Register Gate**Require:** response y , strategy s , seeker profile R , threshold δ

- 1: $r \leftarrow \text{PROFILE}(y)$; $d \leftarrow |h_r - h_s|$
- 2: **if** $d \leq \delta$ **and** $\text{script}_r = \text{script}_s$ **then return** y
- 3: **end if**
- 4: $y' \leftarrow \text{LLM}(\text{rewrite } y \text{ in register } R \text{ keeping } s)$
- 5: $r' \leftarrow \text{PROFILE}(y')$; $d' \leftarrow |h_{r'} - h_s|$
- 6: **if** y' fixes the script, or same script status and $d' < d$ **then return** y'
- 7: **end if**
- 8: **return** y

are kept verbatim. Output parsing is hardened identically for all systems (e.g. markdown emphasis, hyphenated strategy names, hash-seed-dependent strategy order); every deviation from the released code is documented in the repository and applies to every system alike. Every LLM call is cached, so ablations that share stages reuse them.

IV. ESConv-HiEn

We are not aware of a code-mixed ESC test set with strategy labels, so we build one. Each of the 100 test conversations of the base paper is rewritten turn by turn into Roman-script Hinglish by an LLM (gemini-3.5-flash-lite), one call per conversation so that names and spellings stay consistent; strategy labels are copied from the source turns. Two mixing levels are produced: *Light* (target CMI 0.1–0.3) and *Heavy* (0.3–0.5). The CMI of every utterance of at least five words is verified with HingBERT-LID and out-of-band utterances are regenerated (up to three rounds), as are rewrites whose LaBSE [12] similarity to the original drops. An LLM rater (gemma-4-31b-it) scores every conversation for naturalness and meaning preservation (1–5); low-rated conversations are kept unchanged, so the reported numbers include them. A manual rating of a random 20% with the same rubric is prepared (scripts/qc_rater.py) but not yet done. A development set of twelve training conversations is built the same way for tuning δ and selecting the retriever checkpoint. Because the English and Hinglish versions are parallel, any change in system behaviour can be attributed to code-mixing. Table I summarises the data and Table II the quality check. Both versions keep all 3,140 utterances of the 100 conversations. In *Light*, 94.5% of the checked utterances fall in the target band (mean CMI 0.16, 16% Hindi words); in *Heavy*, 87.5% (mean CMI 0.34, half of the words Hindi). The CMI fix rounds rewrote 70% (*Light*) and 48% (*Heavy*) of the utterances at least once, and the mean LaBSE similarity to the English original is 0.88 and 0.71. On the English originals HingBERT-LID tags only 1.3% of the words as Hindi (0.4% after excluding English–Hindi homographs such as *to*, *me*), and 99.6% of the English turns are treated as plain English, so the profiler does not invent code-mixing.

TABLE I

ESConv-HiEn TEST SET. UTTERANCE CMI IS RECOMPUTED WITH THE HINGBERT-LID PROFILER; BAND CHECKS USE UTTERANCES OF AT LEAST FIVE LANGUAGE WORDS (LIGHT: 10–30% HINDI WORDS, HEAVY: CMI 0.3–0.5). SEEKER CMI_s PER TURN IS THE POOLED REGISTER THE AGENTS SEE.

	English	Light	Heavy
Conversations	100	100	100
Utterances	3140	3140	3140
Words / utterance	14.6	15.9	17.0
Mean CMI (all)	0.014	0.162	0.345
Mean CMI (seeker)	0.018	0.158	0.332
Mean CMI (supporter)	0.010	0.166	0.357
Hindi words (%)	1.5	16.3	50.1
In target band (%)	–	94.5	87.5
LaBSE sim. to English	–	0.881	0.709
Regenerated (%)	–	69.9	48.4
Seeker CMI _s per turn	0.017	0.167	0.415
Turns treated as plain English (%)	99.6	8.4	4.3
LID false-Hindi rate (%)	1.26	–	–
... excl. English homographs (%)	0.39	–	–

TABLE II

ESConv-HiEn QUALITY CHECK: MEAN NATURALNESS AND MEANING-PRESERVATION RATINGS (1–5) OF THE HUMAN RATERS (20% SAMPLE) AND OF THE LLM RATER (ALL CONVERSATIONS), THEIR AGREEMENT ON THE OVERLAP (SPEARMAN ρ , QUADRATIC-WEIGHTED κ) AND THE CONVERSATIONS RATED BELOW 3 (REWRITTEN AND CHECKED AGAIN).

Level	Human		LLM		Spearman ρ		Weighted κ		Flagged (<3)
	Nat.	Mean.	Nat.	Mean.	Nat.	Mean.	Nat.	Mean.	
Light	–	–	2.60	4.09	–	–	–	–	49
Heavy	–	–	3.75	4.98	–	–	–	–	0

V. EXPERIMENTAL SETUP

Data. ESConv [2]: the base paper’s test split (the first 100 conversations, 1,210 supporter turns) and case bank (the remaining 1,200 conversations); ESConv-HiEn *Light* and *Heavy*. Multi-agent systems are evaluated on a fixed random sample of 200 supporter turns per version (the same turns in every version), single-call baselines on all turns.

Systems. Zero-shot and few-shot chain-of-thought prompting; MultiAgentESC; a translate-pivot baseline (Hinglish→English translation of the context, MultiAgentESC, English→Hinglish translation of the response into the seeker’s register); CodeMixESC. Ablations on ESConv-HiEn remove the retriever fine-tuning (multilingual encoder as is), the whole cross-lingual retriever (English encoder) and the Register Gate.

Retriever training. paraphrase-multilingual-mpnet-base-v2 (278M parameters) is fine-tuned for one epoch on the 16,951 training pairs (12,156 PHINC, 4,795 ESConv rewrites) with the Multiple Negatives Ranking loss (scale 20, i.e. $\tau = 0.05$), batch size 32 drawn without in-batch duplicates, AdamW with learning rate 2×10^{-5} , 10% warm-up and fp16. The word-embedding matrix (192M parameters) is frozen so that training fits a 4 GB laptop GPU (peak 2.1 GB). The checkpoint is selected every quarter epoch by the mean of dev Overlap@10 and P@10 on the twelve development conversations, whose case-bank entries are excluded; the best is step 264 of 530.

TABLE III

RETRIEVAL ROBUSTNESS ON ESConv-HiEn (ALL TEST TURNS; 1210 QUERIES PER VERSION; $k = 10$; CASE BANK OF 13,484 POSTS). P@10: PROBLEM-TYPE PRECISION OF THE RETRIEVED CASES. OVERLAP@10 (SELF): SHARE OF THE TOP-10 CASES OF THE ENGLISH ORIGINAL THAT THE SAME ENCODER ALSO RETRIEVES FOR THE HINGLISH POST. VALUES IN % ARE MEANS OVER THE QUERIES; THE 95% CLUSTER-BOOTSTRAP INTERVALS (CONVERSATIONS RESAMPLED) HAVE HALF-WIDTHS OF AT MOST 2.4 POINTS. BOLD: BEST ENCODER PER COLUMN; $\dagger$: DIFFERS FROM ROBERTA-L (PAIRED BOOTSTRAP, 95%). ROBERTA-L = ALL-ROBERTA-LARGE-V1, THE ENGLISH RETRIEVER OF MULTIAGENTESC; MPNET = PARAPHRASE-MULTILINGUAL-MPNET-BASE-V2, FT = OUR CONTRASTIVE (MNRL) FINE-TUNING ON HINGLISH-ENGLISH PAIRS.

Encoder	P@10 $\uparrow$			Overlap@10 (self) $\uparrow$	
	EN	Light	Heavy	Light	Heavy
RoBERTa-L (base)	36.0	35.8	33.8	56.9	27.0
LaBSE	29.2 $\dagger$	28.9 $\dagger$	28.5 $\dagger$	61.4 $\dagger$	33.2 $\dagger$
mMPNet	32.8 $\dagger$	31.6 $\dagger$	28.4 $\dagger$	69.9 $\dagger$	31.3 $\dagger$
mMPNet-FT (ours)	32.1 $\dagger$	32.0 $\dagger$	31.5 $\dagger$	73.2$\dagger$	53.3$\dagger$
Random (chance)	20.2	20.2	20.2	–	–

Validation pair accuracy@1 rises from 0.51 to 0.88.

Agents and threshold. All agents of all systems run on gemma-4-26b-a4b-it at temperature 0; the pairwise judge uses the larger gemma-4-31b-it. The Register Gate threshold, tuned on the development set over $\delta \in \{0.05, \dots, 0.35\}$, is $\delta = 0.15$: the largest value whose pooled register distance is within 0.01 of the best and whose chrF does not drop.

Metrics. Retrieval: Overlap@10 between the cases retrieved for a Hinglish query and for its parallel English original, and problem-type Precision@10. Response quality: Distinct-1/2 [16], BLEU-1/2/3 [17], unigram F1, ROUGE-L [18], chrF [19] and multilingual BERTScore [20]. Register: mean CMI gap $|CMI_r - CMI_s|$ and script consistency. Strategy stability: Jensen–Shannon divergence [21] between the strategy distributions on the English and Hinglish versions of the same turns, with a permutation test that swaps each turn’s English and Hinglish labels (the plug-in divergence is biased upwards on finite samples). Pairwise LLM judgement (win/tie/lose, both presentation orders, a win only if both orders agree) on Fluency, Identification, Comforting, Suggestion, Overall and Language Naturalness, complemented by bilingual volunteers on a blind sheet where available. Efficiency: LLM calls and latency per turn. All systems of a table are compared on the same turns, those every system answered; differences are tested with a paired bootstrap that resamples whole conversations, since turns of one conversation are not independent.

VI. RESULTS

A. Retrieval robustness

Code-mixing degrades the base paper’s English retriever: for Heavy posts only 27.0% of its top-10 cases are those it retrieves for the English original (56.9% for Light). LaBSE and multilingual mpnet without fine-tuning help on Light but stay near 31–33% on Heavy, while the contrastively fine-tuned

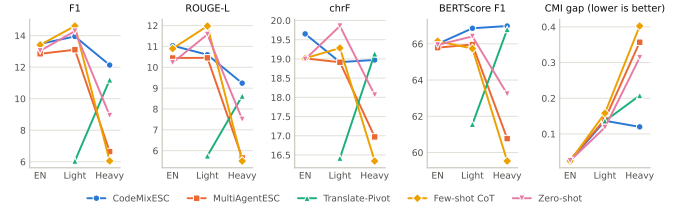


Fig. 2. Response quality and register match from English to Light to Heavy mixing.

encoder keeps 53.3% (Light 73.2%), the largest gain of all encoders, and the strategy distribution of its retrieved cases is the most stable across languages. The price is a lower problem-type precision than the English encoder (31.5 vs 33.8 on Heavy): the fine-tuned space favours the same cases as the English query over the same problem label.

B. Response quality

On Heavy, CodeMixESC reaches ROUGE-L 9.24 and chrF 18.97, above MultiAgentESC (5.66 / 16.97) and above the translate-pivot (8.62 / 19.13); the single-call baselines reach 7.50 (zero-shot) and 5.50 (few-shot CoT). On Light, CodeMixESC reaches ROUGE-L 10.60 and chrF 18.92, on par with MultiAgentESC (10.45 / 18.91) and above the translate-pivot (5.75 / 16.42); the single-call baselines reach 11.58 (zero-shot) and 11.98 (few-shot CoT). On EN, CodeMixESC reaches ROUGE-L 11.04 and chrF 19.65, above MultiAgentESC (10.44 / 19.01); the single-call baselines reach 10.22 (zero-shot) and 10.90 (few-shot CoT). Paired bootstrap over conversations (Table VII): Heavy vs MultiAgentESC: ROUGE-L +3.58 [+1.83, +5.27], $p < 0.01$; Light vs MultiAgentESC: ROUGE-L +0.15 [-1.75, +2.09], $p = 0.93$; Heavy vs Translate-Pivot: ROUGE-L +0.62 [-1.09, +2.40], $p = 0.47$; Light vs Translate-Pivot: ROUGE-L +4.85 [+3.03, +6.79], $p < 0.01$.

C. Register match

Register match (lower is better): on Light the mean CMI gap is 0.137 for CodeMixESC, 0.141 for MultiAgentESC, 0.136 for the pivot and 0.148 before the Register Gate; on Heavy the mean CMI gap is 0.120 for CodeMixESC, 0.357 for MultiAgentESC, 0.208 for the pivot and 0.136 before the Register Gate. The gate’s own effect on the CMI gap (with minus without): Light -0.012 [-0.022, -0.003], $p < 0.01$; Heavy -0.017 [-0.033, -0.001], $p = 0.04$.

D. Strategy stability

Jensen–Shannon divergence between the strategy distributions on the English and the Hinglish versions of the same turns (lower = less disturbed by code-mixing): CodeMixESC 0.022 (Light) / 0.016 (Heavy); MultiAgentESC 0.014 (Light) / 0.081 (Heavy); Translate-Pivot 0.015 (Light) / 0.016 (Heavy).

TABLE IV

RESPONSE QUALITY ON HEAVY (50 COMMON TURNS OF THE 50-TURN SUBSET). DISTINCT-N (D), CORPUS BLEU-N (B), UNIGRAM F1, ROUGE-L (R-L), CORPUS CHRf AND MULTILINGUAL BERTSCORE F1, ALL $\times 100$; BEST PER COLUMN IN BOLD.

System	D-1	D-2	B-1	B-2	B-3	F1	R-L	chrF	BERTScore
Zero-shot	34.15	71.48	13.20	4.52	2.19	8.93	7.50	18.06	63.23
Few-shot CoT	30.56	69.49	9.94	2.63	1.18	6.05	5.50	16.34	59.53
MultiAgentESC	35.66	73.35	10.59	3.05	1.28	6.65	5.66	16.97	60.77
Translate-Pivot	31.79	71.08	15.09	5.00	1.99	11.19	8.62	19.13	66.79
CodeMixESC	32.86	75.00	15.28	4.77	1.92	12.13	9.24	18.97	66.98

TABLE V

RESPONSE QUALITY ON LIGHT (50 COMMON TURNS OF THE 50-TURN SUBSET). DISTINCT-N (D), CORPUS BLEU-N (B), UNIGRAM F1, ROUGE-L (R-L), CORPUS CHRf AND MULTILINGUAL BERTSCORE F1, ALL $\times 100$; BEST PER COLUMN IN BOLD.

System	D-1	D-2	B-1	B-2	B-3	F1	R-L	chrF	BERTScore
Zero-shot	26.08	61.21	19.08	7.70	3.60	14.28	11.58	19.86	66.41
Few-shot CoT	27.23	66.41	19.44	6.81	2.96	14.61	11.98	19.28	65.74
MultiAgentESC	29.83	67.14	18.03	6.12	2.05	13.10	10.45	18.91	65.96
Translate-Pivot	32.38	69.21	11.07	3.23	1.33	6.04	5.75	16.42	61.56
CodeMixESC	30.76	68.91	19.28	7.01	3.13	13.95	10.60	18.92	66.86

TABLE VI

RESPONSE QUALITY ON EN (50 COMMON TURNS OF THE 50-TURN SUBSET). DISTINCT-N (D), CORPUS BLEU-N (B), UNIGRAM F1, ROUGE-L (R-L), CORPUS CHRf AND MULTILINGUAL BERTSCORE F1, ALL $\times 100$; BEST PER COLUMN IN BOLD.

System	D-1	D-2	B-1	B-2	B-3	F1	R-L	chrF	BERTScore
Zero-shot	28.78	68.38	15.61	3.83	1.49	13.04	10.22	18.99	65.94
Few-shot CoT	30.36	68.92	15.82	3.86	1.35	13.41	10.90	19.02	66.15
MultiAgentESC	31.58	70.08	16.17	3.55	1.27	12.85	10.44	19.01	65.79
CodeMixESC	30.70	70.77	16.46	4.57	2.01	13.48	11.04	19.65	65.99

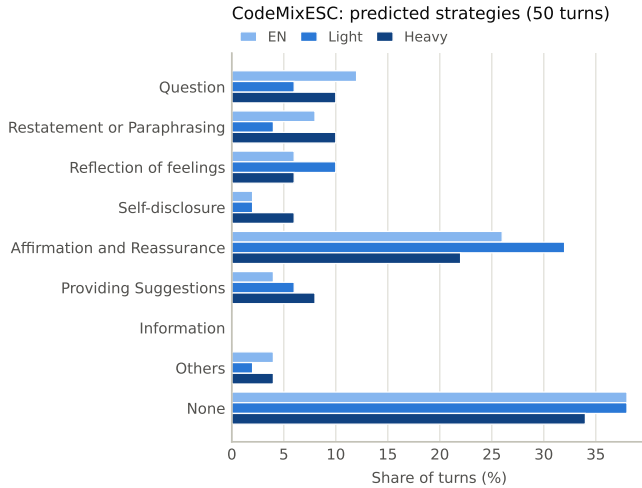


Fig. 3. Strategy distribution of CodeMixESC on the English, Light and Heavy versions.

E. Ablations

On Heavy, CodeMixESC: ROUGE-L 9.24, CMI gap 0.120; w/o Register Gate: ROUGE-L 9.10, CMI gap 0.136; w/o retriever fine-tuning: ROUGE-L 9.25, CMI gap 0.116; w/o cross-lingual retriever: ROUGE-L 10.44, CMI gap 0.099. Each row removes one component from the full system on the same

turns, so the differences isolate the retriever fine-tuning, the cross-lingual retriever and the Register Gate. Reply quality does not depend measurably on the retriever: with the English encoder instead of the cross-lingual one (ROUGE-L difference -1.20 [-2.71, +0.08], $p = 0.07$) or without the fine-tuning (ROUGE-L difference -0.01 [-1.08, +1.12], $p = 0.97$) the ROUGE-L difference to the full system is not significant, so the gain of CodeMixESC over MultiAgentESC on Heavy comes from the register-aware agents rather than from the retriever (the Register Gate adds a small improvement of the CMI gap, not of ROUGE-L). The fine-tuned retriever contributes the robust retrieval itself (Table III), which on this 50-turn sample does not translate into better replies; this is a limitation of the evaluation as much as of the retriever.

F. Pairwise judgement

Pairwise judgement by gemma-4-31b-it in both presentation orders (a win only if both orders agree): CodeMixESC vs MultiAgentESC on Light: overall win/tie/lose 74/6/20 percent of the turns ($p < 0.01$); CodeMixESC vs Translate-Pivot on Light: overall win/tie/lose 44/24/32 percent of the turns ($p = 0.42$); CodeMixESC vs MultiAgentESC on Heavy: overall win/tie/lose 86/10/4 percent of the turns ($p < 0.01$); CodeMixESC vs Translate-Pivot on Heavy: overall win/tie/lose 56/20/24 percent of the turns ($p = 0.02$).

TABLE VII

PAIRED BOOTSTRAP (2000 RESAMPLES, SEED 42) OF THE PER-TURN SCORE DIFFERENCES $\Delta = \text{CodeMixESC} - \text{SYSTEM}$ ON THE COMMON TURNS (SENTENCE CHRf; CMI GAP: LOWER IS BETTER). [†] $p < 0.05$, [‡] $p < 0.01$ (TWO-SIDED); THE MARKDOWN AND CSV VERSIONS GIVE THE 95% CIs.

Version	CodeMixESC	n	F1	ROUGE-L	chrF	BERTScore	CMI gap
EN	vs Zero-shot	50	+0.44	+0.82	+0.27	+0.06	−0.001
	vs Few-shot CoT	50	+0.07	+0.14	+0.32	−0.15	+0.001
	vs MultiAgentESC	50	+0.63	+0.59	+0.57	+0.20	−0.003
Light	vs Zero-shot	50	−0.34	−0.97	−0.44	+0.45	+0.019
	vs Few-shot CoT	50	−0.66	−1.38	−0.09	+1.12 [‡]	−0.021
	vs MultiAgentESC	50	+0.84	+0.15	+0.29	+0.90 [†]	−0.004
	vs Translate-Pivot	50	+7.90 [‡]	+4.85 [‡]	+2.62 [‡]	+5.29 [‡]	+0.001
	vs w/o Register Gate	50	+0.22	+0.03	+0.04	+0.31 [†]	−0.012 [‡]
Heavy	vs Zero-shot	50	+3.20 [†]	+1.74	+1.31 [†]	+3.75 [‡]	−0.194 [‡]
	vs Few-shot CoT	50	+6.08 [‡]	+3.74 [‡]	+2.43 [‡]	+7.45 [‡]	−0.282 [‡]
	vs MultiAgentESC	50	+5.49 [‡]	+3.58 [‡]	+2.06 [‡]	+6.21 [‡]	−0.237 [‡]
	vs Translate-Pivot	50	+0.94	+0.62	+0.15	+0.19	−0.088 [‡]
	vs w/o Register Gate	50	+0.27	+0.14	+0.04	+0.13	−0.017 [†]
	vs w/o retriever fine-tuning	50	+0.38	−0.01	+0.02	−0.06	+0.004
	vs w/o cross-lingual retriever	50	−0.52	−1.20	−0.04	−0.83 [†]	+0.021

TABLE VIII

REGISTER MATCH OF THE RESPONSES TO THE SEEKER ON THE COMMON TURNS: MEAN CMI GAP $|\text{CMI}_r - \text{CMI}_s|$ AND HINDI-FRACTION GAP (LOWER IS BETTER), AND SCRIPT CONSISTENCY IN % (HIGHER IS BETTER). MEAN SEEKER CMI_s: EN 0.019, LIGHT 0.181, HEAVY 0.420. GOLD REFERENCE: THE DATASET’S OWN SUPPORTER TURNS. PROFILER: HINGBERT-LID.

System	EN			Light			Heavy		
	CMI gap ↓	Hi gap ↓	Script ↑	CMI gap ↓	Hi gap ↓	Script ↑	CMI gap ↓	Hi gap ↓	Script ↑
Zero-shot	0.025	0.025	100.0	0.118	0.130	100.0	0.314	0.378	100.0
Few-shot CoT	0.024	0.024	100.0	0.158	0.158	100.0	0.402	0.515	100.0
MultiAgentESC	0.028	0.028	100.0	0.141	0.153	100.0	0.357	0.452	100.0
Translate-Pivot	—	—	—	0.136	0.521	100.0	0.208	0.223	100.0
CodeMixESC	0.025	0.025	100.0	0.137	0.144	100.0	0.120	0.135	100.0
w/o Register Gate	—	—	—	0.148	0.167	100.0	0.136	0.175	100.0
w/o retriever fine-tuning	—	—	—	—	—	—	0.116	0.129	100.0
w/o cross-lingual retriever	—	—	—	—	—	—	0.099	0.115	100.0
Gold reference	0.026	0.026	100.0	0.063	0.070	100.0	0.076	0.110	100.0

TABLE IX

STRATEGY STABILITY UNDER CODE-MIXING ON THE COMMON TURNS: JENSEN-SHANNON DIVERGENCE (BITS) BETWEEN THE PREDICTED-STRATEGY DISTRIBUTIONS OF EN AND OF THE HINGLISH VERSION, OVER THE 8 STRATEGIES + NONE (JSD) AND OVER THE 8 STRATEGIES ON TURNS WHERE BOTH VERSIONS PREDICTED ONE (JSD-S), AND PER-TURN AGREEMENT IN %. [†] COMPARED WITH MULTIAGENTESC ON EN (THE PIVOT RUNS MULTIAGENTESC ON TRANSLATIONS).

System	EN→Light			EN→Heavy		
	JSD ↓	JSD-S ↓	Agr. ↑	JSD ↓	JSD-S ↓	Agr. ↑
Few-shot CoT	0.004	0.004	86.0	0.005	0.005	86.0
MultiAgentESC	0.014	0.023	60.0	0.081	0.122	54.0
Translate-Pivot [†]	0.015	0.038	66.0	0.016	0.022	66.0
CodeMixESC	0.022	0.029	70.0	0.016	0.039	62.0

G. Efficiency

CodeMixESC needs 10.20 LLM calls per turn on Light vs 9.66 for MultiAgentESC and 11.72 for the pivot; 10.70 LLM calls per turn on Heavy vs 10.22 for MultiAgentESC and 11.22 for the pivot: the profiler and the retriever add no LLM call, and the Register Gate at most one.

H. Case studies

Cases are selected automatically by fixed criteria (`scripts/case_studies.py`): register rescues, Register Gate interventions, strategy divergences and failures.

The cases are selected by fixed rules (register rescues, gate rewrites, strategy changes and failures), so they show both what the components fix and where the system still misses the seeker’s register.

TABLE X
ABLATIONS OF CODEMIXESC ON THE COMMON TURNS. QUALITY SCORES $\times 100$; BEST PER VERSION IN BOLD. W/O REGISTER GATE IS THE SAME RUN BEFORE THE GATE.

Version	System	D-2	B-2	F1	R-L	chrF	BERTScore	CMI gap $\downarrow$	Script $\uparrow$
Light	CodeMixESC	68.91	7.01	13.95	10.60	18.92	66.86	0.137	100.0
	w/o Register Gate	69.49	6.87	13.73	10.58	18.88	66.54	0.148	100.0
Heavy	CodeMixESC	75.00	4.77	12.13	9.24	18.97	66.98	0.120	100.0
	w/o Register Gate	75.30	4.73	11.86	9.10	18.92	66.86	0.136	100.0
	w/o retriever fine-tuning	70.90	4.83	11.75	9.25	18.96	67.04	0.116	100.0
	w/o cross-lingual retriever	73.93	5.44	12.65	10.44	18.90	67.81	0.099	100.0

TABLE XI
PAIRWISE LLM JUDGE (GEMMA-4-31B-IT, BOTH RESPONSE ORDERS) ON A SEEDED SAMPLE OF TURNS. CELLS: WIN / TIE / LOSE IN % OF THE TURNS; $\dagger$ $p < 0.05$, $\ddagger$ $p < 0.01$ (TWO-SIDED SIGN TEST OF WINS AGAINST LOSSES).

Version	Comparison	n	Fluency	Identification	Comforting	Suggestion	Overall	Language Naturalness
Light	CodeMixESC vs MultiAgentESC	50	0 / 100 / 0	16 / 70 / 14	60 / 18 / 22 $\ddagger$	6 / 86 / 8	74 / 6 / 20 $\ddagger$	76 / 18 / 6 $\ddagger$
	CodeMixESC vs Translate-Pivot	50	0 / 100 / 0	8 / 78 / 14	44 / 24 / 32	4 / 88 / 8	44 / 24 / 32	46 / 40 / 14 $\ddagger$
Heavy	CodeMixESC vs MultiAgentESC	50	2 / 98 / 0	6 / 90 / 4	64 / 28 / 8 $\ddagger$	8 / 90 / 2	86 / 10 / 4 $\ddagger$	82 / 16 / 2 $\ddagger$
	CodeMixESC vs Translate-Pivot	50	4 / 96 / 0	14 / 74 / 12	42 / 30 / 28	8 / 80 / 12	56 / 20 / 24 $\dagger$	50 / 44 / 6 $\ddagger$

TABLE XII
EFFICIENCY ON THE COMMON TURNS: LOGICAL LLM CALLS AND SUMMED API LATENCY PER TURN, EXTRA CALLS OVER MULTIAGENTESC ON THE SAME TURNS, SHARE OF TURNS ON THE FULL MULTI-AGENT PATH, AND RESPONSE LENGTH. CALLS COUNT CACHED CALLS TOO; LATENCY IS THE API TIME OF THE ORIGINAL CALLS.

Version	System	Calls/turn	Δ calls	Latency (s)	Multi %	Words	>30 w. %
EN	Zero-shot	1.00	-8.80	2.1	0.0	23.8	0.0
	Few-shot CoT	1.00	-8.80	3.1	0.0	22.2	0.0
	MultiAgentESC	9.80	+0.00	49.6	62.0	22.3	0.0
	CodeMixESC	9.74	-0.06	49.5	62.0	23.1	4.0
Light	Zero-shot	1.00	-8.66	2.0	0.0	23.1	0.0
	Few-shot CoT	1.00	-8.66	3.0	0.0	22.3	0.0
	MultiAgentESC	9.66	+0.00	60.3	62.0	22.5	0.0
	Translate-Pivot	11.72	+2.06	67.1	62.0	24.4	0.0
	CodeMixESC	10.20	+0.54	71.9	62.0	22.2	0.0
	w/o Register Gate	9.72	+0.06	70.9	62.0	22.3	0.0
Heavy	Zero-shot	1.00	-9.22	2.1	0.0	23.7	0.0
	Few-shot CoT	1.00	-9.22	2.9	0.0	22.7	0.0
	MultiAgentESC	10.22	+0.00	68.0	66.0	23.2	0.0
	Translate-Pivot	11.22	+1.00	62.2	58.0	24.3	0.0
	CodeMixESC	10.70	+0.48	61.0	66.0	23.8	0.0
	w/o Register Gate	10.14	-0.08	59.8	66.0	23.8	0.0
	w/o retriever fine-tuning	10.38	+0.16	56.4	64.0	23.9	2.0
	w/o cross-lingual retriever	10.60	+0.38	58.3	66.0	23.1	0.0

TABLE XIII
CASE 1 (RESCUE, LIGHT, SEEKER CMI 0.18).

Supporter	What sort of things are you trying to decide, yaar?
Seeker	Pata nahi, I don't even know where to start.
Seeker	I'm trying to decide, yaar, if I can build trust with him again, sach mein.
Seeker	He lied about everything.
Reference	Well, let's try to take it one problem at a time so's not to get overwhelmed, yaar. What is your biggest problem at the moment? Ah, sach mein am I to take it that a relationship has recently ended, yaar?
MultiAgentESC [Question]	That sounds incredibly painful; can you tell me more about how his lies have affected your sense of security?
Pivot [Question]	Yeh sunkar bahut bura laga, yaar. Jab tum trust rebuild karne ka sochte ho, toh abhi sabse badi cheez kya tumhe rok rahi hai?
CodeMixESC [Affirmation and Reassurance]	I can only imagine how much this hurts. It is completely natural to feel lost and betrayed right now, yaar. I am here for you.

TABLE XIV
CASE 2 (RESCUE, LIGHT, SEEKER CMI 0.16).

Seeker	I really appreciate you listening yaar, and understanding.
Seeker	it's tremendously helpful yaar
Supporter	Theek hai, you are welcome. Everything will be okay bas just know that.
Seeker	You're right yaar...I do know that, bas I just need to truly believe it. Every day is a new opportunity.
Reference	Right, aur you have to take control of those negative thoughts, bas replace them with positive ones.
MultiAgentESC [Reflection of feelings]	It sounds like you are feeling a sense of hope, yet you're also being realistic about the effort it takes to truly believe in it.
Pivot [Reflection of feelings]	Aisa lag raha hai ki aap future ko lekar hopeful toh ho, par shayad pooraa trust nahi kar paa rahe ki sab theek ho jayega.
CodeMixESC [Affirmation and Reassurance]	Bilkul sahi kaha yaar. It takes time, par you are already taking the right steps. Just keep going, you will get through this.

TABLE XV
CASE 3 (GATE, LIGHT, SEEKER CMI 0.17).

Seeker	What kind of jobs did you take, Tom bhai?
Supporter	Work part-time and go to school part-time yaar.
Seeker	I'm trying to find a steady job, koshish jaari hai.
Seeker	Jobs that can help support both my family aur my tuition yaar.
Reference	I suggest creating flyers and ask to be a dog walker, sach mein. Americans love their dogs bahut. 70 percent of Americans have dogs.
MultiAgentESC [Question]	I understand, that is a lot to carry. What kind of roles or industries are you currently looking into to balance your studies?
Pivot [Reflection of feelings]	Lagta hai in responsibilities ki wajah se aap bahut pressured feel kar rahe ho, phir bhi aap is struggle se nikalne ke liye committed ho.
CodeMixESC (pre-gate)	So, you are looking for a stable job jo aapki family ki help bhi kare aur aapki tuition fees bhi manage kar sake?
CodeMixESC [Restatement or Paraphrasing]	So, you are looking for a steady job jo aapki family ko support kare aur aapki tuition fees bhi manage kar sake?

TABLE XVI
CASE 4 (GATE, LIGHT, SEEKER CMI 0.15).

Supporter	Anything else I could help you with today, theek hai?
Seeker	No, theek hai, I'll try following the advice you suggested. Hopefully, that'll help me get out of my slump.
Seeker	I may even meet another dog that way too, theek hai.
Seeker	Thank you again, yaar.
Reference	Thank you yaar. Please use "Finish and Quit" to end the discussion, theek hai.
MultiAgentESC [Information]	You're very welcome, yaar. Take it one step at a time. I'm sure you'll find peace through your wonderful work with pets.
Pivot [Reflection of feelings]	You're welcome. Lagta hai aap thoda zyada hopeful feel kar rahe ho aur in positive steps ke liye ready ho.
CodeMixESC (pre-gate)	You are most welcome, yaar. Do you think aap online advocacy se start karoge ya shelter volunteering se?
CodeMixESC [Question]	You are most welcome, yaar. Do you think you will start with online advocacy or shelter volunteering?

TABLE XVII
CASE 5 (STRATEGY, LIGHT, SEEKER CMI 0.20).

Supporter	Could you perhaps set up Zoom meetings where you could play together online, theek rahega?
Seeker	Any idea yaar ki kaise yeh work karega? Would there be any delay between the sounds?
Supporter	No I don't think so :) Mujhe lagta hai that it is totally live toh you could play just like normal.
Seeker	Thanks for the idea yaar, sach mein I will give it a try.
Reference	I hope it goes well for you. It has been such an awful year, hai na?
MultiAgentESC [Question]	You're most welcome! Do you think your band members would be open to trying a virtual jam session?
Pivot [Question]	You're welcome! Kya aapko lagta hai ki aapke band members virtual jam session try karne ke liye open honge?
CodeMixESC [Others]	You are very welcome! I really hope it works out for you and your band. All the best!

TABLE XVIII
CASE 6 (STRATEGY, LIGHT, SEEKER CMI 0.15).

Supporter	With him, ya a relationship with someone else?
Seeker	With him... he was perfect, but also, I'm lonely yaar, and it's bahut hard to process not being in a relationship during the pandemic.
Supporter	Have you tried making other connections online? Bahut saare people are ghar pe during this pandemic, so you may meet someone much better sach mein.
Seeker	True yaar. What can I do to get over him in the meantime, batao?
Reference	Keep busy yaar! Get mad instead of sad! I would not contact him if he was not contacting me yaar. You deserve better than that!
MultiAgentESC [Information]	Try practicing mindfulness, limiting social media to avoid seeing him online, and picking up new hobbies to redirect your energy and ease the loneliness.
Pivot [Information]	Aap mindfulness try kar sakte ho, journaling kar lo, ya phir digital boundaries set karo taaki uska online presence kam dekhna pade.
CodeMixESC [Providing Suggestions]	Try focusing on hobbies or journaling your feelings. Also, limiting his social media stalking will help you heal faster, sach mein.

TABLE XIX
CASE 7 (FAILURE, LIGHT, SEEKER CMI 0.17).

Seeker	What kind of jobs did you take, Tom bhai?
Supporter	Work part-time and go to school part-time yaar.
Seeker	I'm trying to find a steady job, koshish jaari hai.
Seeker	Jobs that can help support both my family aur my tuition yaar.
Reference	I suggest creating flyers and ask to be a dog walker, sach mein. Americans love their dogs bahut. 70 percent of Americans have dogs.
MultiAgentESC [Question]	I understand, that is a lot to carry. What kind of roles or industries are you currently looking into to balance your studies?
Pivot [Reflection of feelings]	Lagta hai in responsibilities ki wajah se aap bahut pressured feel kar rahe ho, phir bhi aap is struggle se nikalne ke liye committed ho.
CodeMixESC [Restatement or Paraphrasing]	So, you are looking for a steady job jo aapki family ko support kare aur aapki tuition fees bhi manage kar sake?

TABLE XX
CASE 8 (FAILURE, LIGHT, SEEKER CMI 0.17).

Seeker	That might be a good idea, sach mein.
Seeker	Do you have any recommendations, yaar?
Supporter	I know kitna difficult hai yaar - I have been on furlough too and am finding financial things bahut tricky.
Seeker	I guess sach mein we're in the same boat.
Reference	Bahut sare log sach mein suffer kar rahe hain yaar at the moment. It's been such a terrible year but I am hopeful things will get better.
MultiAgentESC [Affirmation and Reassurance]	Haan yaar, bilkul. It is tough, but knowing we are not alone makes it a little easier. We can get through this.
Pivot [Question]	Hum ye challenges saath mein handle kar rahe hain, toh kya aap un online gig work ideas ke baare mein aur baat karna chahenge jo humne pehle discuss kiye the?
CodeMixESC [Affirmation and Reassurance]	Haan yaar, bilkul. Don't worry, hum rasta nikal lenge. You are strong and I know you will manage it well.

TABLE XXI
CASE 9 (RESCUE, HEAVY, SEEKER CMI 0.39).

Seeker	thats great aur nahi, snowfall nahi ho rahi hai, but cold bahut zyada hai
Seeker	merry christmas aapko bhi
Supporter	How can I help you today, kya main help karun?
Seeker	mujhe kuch issues ho rahe hain friends ke sath jo actually friends nahi hain
Reference	I hear you, aapko samajhne mein trouble ho rahi hai ki kaunse friends sach mein aapke friends hain aur kaunse nahi. Is that about right?
MultiAgentESC [Reflection of feelings]	It sounds like you're feeling hurt and disappointed because the people you trusted as friends aren't acting like true friends. Do you want to share more?
Pivot [None]	Sun kar bahut bura laga. Jab log waise nahi nikalte jaisa humne socha tha, toh bahut dard hota hai. Kya aap mujhe aur batana chahenge?
CodeMixESC [Question]	I am sorry to hear that. Kya aap batana chahenge ki exactly kya hua jis wajah se aapko aisa feel ho raha hai?

TABLE XXII
CASE 10 (RESCUE, HEAVY, SEEKER CMI 0.40).

Seeker	Aapne kis type ki jobs li thi honestly?
Supporter	Part-time work karo aur part-time hi school jao.
Seeker	Main ek steady job find karne ki try kar raha hoon.
Seeker	Aisi jobs jo meri family ko bhi support karein aur meri tuition fees mein bhi help karein.
Reference	Mera suggestion hai ki flyers create karo aur dog walker banne ke liye ask karo. Americans ko apne dogs se bahut pyaar hota hai. 70 percent Americans ke paas dogs hote hain.
MultiAgentESC [Affirmation and Reassurance]	It is incredibly brave of you to step up for your family. You are handling a heavy burden, and we will work through this together.
Pivot [Providing Suggestions]	Kya aapne campus mein work-study programs ya remote part-time jobs try kiye hain? Isse aapko flexibility bhi mil jayegi aur help bhi hogi.
CodeMixESC [Restatement or Paraphrasing]	I understand. Aap ek aisi steady job dhoond rahe ho jo aapki family ko support kare aur saath hi aapki tuition fees bhi manage kar sake.

TABLE XXIII
CASE 11 (GATE, HEAVY, SEEKER CMI 0.46).

Supporter	Tum kis sort ki cheezein decide karne ki try kar rahi ho?
Seeker	I don't even know ki kahan se start karu main.
Seeker	Main yeh decide karne ki try kar rahi hu ki I can build trust with him again or not.
Seeker	He lied about everything.
Reference	Well, let's try to take it one problem at a time taaki tum overwhelmed na ho jao. What is your biggest problem currently? Ah, kya main yeh assume karu ki koi relationship recently end hua hai?
MultiAgentESC [Restatement or Paraphrasing]	I hear you. It sounds like you're feeling deeply hurt and uncertain because his lies have completely broken the trust you once had.
Pivot [None]	Yeh sunkar bahut bura laga, it sounds so overwhelming. Jhoot sunna bahut painful hota hai. Kya aap comfortable ho batane mein ki exactly kya hua?
CodeMixESC (pre-gate)	I understand. Uska har baat par jhoot bolna tumhare liye bahut bada betrayal hai, aur ab tum confused ho ki trust phir se build kar paogi ya nahi.
CodeMixESC [Restatement or Paraphrasing]	I understand. Uska har baat par jhoot bolna ek bahut bada betrayal hai, isliye tum confused ho ki trust phir se build kar paogi ya nahi.

TABLE XXIV
CASE 12 (GATE, HEAVY, SEEKER CMI 0.44).

Supporter	Anything else main help kar sakta hoon aaj?
Seeker	Nahi, I will try following your advice. Hopefully, usse mujhe is slump se bahar aane mein help milegi.
Seeker	Maybe is process mein mujhe koi aur dog bhi mil jaye.
Seeker	Thank you again.
Reference	Thank you. Discussion end karne ke liye please "Finish and Quit" use karo.
MultiAgentESC [Reflection of feelings]	You're very welcome. It sounds like you're feeling a sense of hope and purpose now that you have a plan to honor your dog's memory.
Pivot [Reflection of feelings]	You're welcome. Aisa lag raha hai ki aap hopeful ho aur apne grief ko doosre animals ke liye kuch meaningful banane ke liye motivated ho.
CodeMixESC (pre-gate)	You're most welcome. I'm sure ye process tumhe bahut help karega. Take it one step at a time, sab theek ho jayega.
CodeMixESC [Affirmation and Reassurance]	You're most welcome. I'm sure ye process tumhe bahut help karega. Bas ek ek step karke aage badho, sab theek ho jayega.

TABLE XXV
CASE 13 (STRATEGY, HEAVY, SEEKER CMI 0.46).

Supporter	Good hoon, I'm ok thanks. Aaj kya discuss karna hai?
Seeker	I am feeling depressed ghar pe reh kar due to COVID.
Supporter	Oh sun kar bad feel hua. Would you feel thoda happy agar yeh socho ki jab yeh over hoga, next year shayad?
Seeker	Absolutely... I just can't wait jab tak yeh over ho jaye !!
Reference	I know right! Fir aap kya karoge jab yeh completely over ho jayega!?
MultiAgentESC [Others]	I can feel your excitement! What is the first thing you want to do once everything returns to normal?
Pivot [Question]	Main tumhara excitement samajh sakta hoon! Jab tak hum wait kar rahe hain, kya tumne socha hai ki bahar jaane ke baad tum kya choti-choti cheezein karna chahoge?
CodeMixESC [Question]	I can feel that excitement! Toh, jab sab normal ho jayega, toh sabse pehle aap kya karna chahoge?

TABLE XXVI
CASE 14 (STRATEGY, HEAVY, SEEKER CMI 0.43).

Supporter	Mere liye personally online gaming ek great way hai to interact with others aur thoda fun karne ke liye.
Seeker	Maybe main try karunga. Koi game suggestions hain kya?
Supporter	Tum kis type ki movie dekh rahe the? Maybe choose a happier movie instead of a sad one.
Seeker	Main comedy dekh raha tha. It's a movie I usually like, par main apna mind divert nahi kar paya.
Reference	I would recommend games like Hearthstone for strategy, WoW for role-playing, and agar tumhe sci-fi pasand hai toh Cyberpunk abhi recently hi aayi hai. Really koi bhi game jisme multiplayer ho, that would be a very good fit!
MultiAgentESC [Affirmation and Reassurance]	It's completely understandable that a movie isn't enough right now. Don't be hard on yourself; it's really tough to focus when you're feeling this lonely.
Pivot [Providing Suggestions]	Agar passive watching se kaam nahi ban raha, toh maybe koi puzzle ya game try karo? Isse tumhara mind actively engaged rahega.
CodeMixESC [Providing Suggestions]	I understand, kabhi kabhi movies bhi kaam nahi karti jab mind itna heavy ho. Kya tumne kabhi kuch creative ya hands-on try kiya hai, like drawing ya cooking?

TABLE XXVII
CASE 15 (FAILURE, HEAVY, SEEKER CMI 0.46).

Supporter	Tum kis sort ki cheezein decide karne ki try kar rahi ho?
Seeker	I don't even know ki kahan se start karu main.
Seeker	Main yeh decide karne ki try kar rahi hu ki I can build trust with him again or not.
Seeker	He lied about everything.
Reference	Well, let's try to take it one problem at a time taaki tum overwhelmed na ho jao. What is your biggest problem currently? Ah, kya main yeh assume karu ki koi relationship recently end hua hai?
MultiAgentESC [Restatement or Paraphrasing]	I hear you. It sounds like you're feeling deeply hurt and uncertain because his lies have completely broken the trust you once had.
Pivot [None]	Yeh sunkar bahut bura laga, it sounds so overwhelming. Jhoot sunna bahut painful hota hai. Kya aap comfortable ho batane mein ki exactly kya hua?
CodeMixESC [Restatement or Paraphrasing]	I understand. Uska har baat par jhoot bolna ek bahut bada betrayal hai, isliye tum confused ho ki trust phir se build kar paogi ya nahi.

TABLE XXVIII
CASE 16 (FAILURE, HEAVY, SEEKER CMI 0.48).

Seeker	Par is situation mein what can I do with them?
Seeker	Tumhare paas koi idea hai kya?
Supporter	Simple sa kuch time sath mein spend karna helpful ho sakta hai. Connecting with someone who cares about you can help reassure your positive feelings.
Seeker	Ohhh... I understand
Reference	Agar tum comfortable ho, toh kuch outdoors karna accha rahega. Fresh air aur physical activity anxiety ko kam karne mein help kar sakti hai aur koi mutual activity karne se logon ke beech bond strong hota hai.
MultiAgentESC	That's good! Maybe you could start by just grabbing a coffee or watching a movie with them to keep things light and easy?
[Providing Suggestions]	
Pivot [Reflection of feelings]	I can sense ki ab aapko thoda relief aur clarity feel ho rahi hai, ye dekh kar ki unke saath rehna aapko kaise help kar sakta hai.
CodeMixESC	Glad you got it! Toh kya tum aaj ya kal mein apne us well-wisher se milne ya baat karne ka plan bana sakte ho?
[Question]	

VII. LIMITATIONS AND ETHICS

The work covers single-session, text-only, Roman-script Hindi–English conversations with the ESConv strategies; other Indian languages, speech and crisis counselling are out of scope. ESConv-HiEn is a machine rewrite of English conversations: although CMI-verified and quality-checked, it is not spontaneous Hinglish, and its references inherit the LLM’s mixing style. Its quality is rated by an LLM only: meaning is well preserved (4.09 Light, 4.98 Heavy, of 5), but Light rewrites read less naturally (2.60 of 5, 49 of 100 conversations below 3; Heavy 3.75), with overused fillers such as *yaar* and some ungrammatical mixing. Automatic metrics against a single reference reward surface overlap; the pairwise judgement and the register metrics complement them. The system is a research prototype for emotional support and does not replace professional help; it must not be deployed for crisis situations. ESConv is licensed for academic research only, and ESConv-HiEn is shared under the same terms.

VIII. CONCLUSION

CodeMixESC adapts a multi-agent emotional-support pipeline to code-mixed Hinglish help-seekers without fine-tuning the agents. A contrastively fine-tuned multilingual encoder makes retrieval robust to code-mixing (27% $\rightarrow$ 53% Overlap@10 on Heavy), the register profile and the Register Gate bring the replies closer to the seeker’s language mix at a cost of at most one LLM call, and ESConv-HiEn makes the effect of code-mixing measurable on parallel conversations. Future work: spontaneous (not rewritten) Hinglish, other Indian languages and human evaluation with native speakers.

REFERENCES

- [1] G. Gururaj, M. Varghese, V. Benegal, G. N. Rao, K. Pathak, L. K. Singh *et al.*, “National mental health survey of India, 2015–16: Summary,” National Institute of Mental Health and Neuro Sciences (NIMHANS), Bengaluru, India, Tech. Rep., 2016.
- [2] S. Liu, C. Zheng, O. Demasi, S. Sabour, Y. Li, Z. Yu, Y. Jiang, and M. Huang, “Towards emotional support dialog systems,” in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (ACL-IJCNLP)*, 2021, pp. 3469–3483.
- [3] Y. Xu, J. Hu, Z. Zhao, Z. Duan, X. Sun, and X. Yang, “MultiAgentESC: A LLM-based multi-agent collaboration framework for emotional support conversation,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025, pp. 4665–4681.
- [4] A. Srivastava, K. Mittal, A. Tiwari, and M. S. Akhtar, “Knowledge-infused hierarchy-aware emotion recognition in code-mixed mental health counseling conversations,” in *Proceedings of the Fifteenth Language Resources and Evaluation Conference (LREC)*, 2026, pp. 2887–2898.
- [5] G. I. Winata, D. Anugraha, P. A. Irawan *et al.*, “Can large language models understand, reason about, and generate code-switched text?” *arXiv preprint arXiv:2601.07153*, 2026.
- [6] Q. Wu, G. Bansal, J. Zhang, Y. Wu, B. Li, E. Zhu, L. Jiang, X. Zhang, S. Zhang, J. Liu *et al.*, “AutoGen: Enabling next-gen LLM applications via multi-agent conversation,” in *ICLR 2024 Workshop on Large Language Model (LLM) Agents*, 2024.
- [7] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 3982–3992.

- [8] R. Nayak and R. Joshi, “L3Cube-HingCorpus and HingBERT: A code mixed Hindi-English dataset and BERT language models,” in *Proceedings of the WILDRE-6 Workshop within the 13th Language Resources and Evaluation Conference*, Marseille, France, 2022.
- [9] V. Srivastava and M. Singh, “PHINC: A parallel Hinglish social media code-mixed corpus for machine translation,” in *Proceedings of the Sixth Workshop on Noisy User-generated Text (W-NUT 2020)*, 2020.
- [10] S. Kumar and T. Chakraborty, “Harmonizing code-mixed conversations: Personality-assisted code-mixed response generation in dialogues,” in *Findings of the Association for Computational Linguistics: EACL 2024*, 2024.
- [11] N. Reimers and I. Gurevych, “Making monolingual sentence embeddings multilingual using knowledge distillation,” in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 4512–4525.
- [12] F. Feng, Y. Yang, D. Cer, N. Arivazhagan, and W. Wang, “Language-agnostic BERT sentence embedding,” in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022, pp. 878–891.
- [13] M. Henderson, R. Al-Rfou, B. Strope, Y.-H. Sung, L. Lukacs, R. Guo, S. Kumar, B. Miklos, and R. Kurzweil, “Efficient natural language response suggestion for smart reply,” *arXiv preprint arXiv:1705.00652*, 2017.
- [14] H. Giles, N. Coupland, and J. Coupland, “Accommodation theory: Communication, context, and consequence,” in *Contexts of Accommodation: Developments in Applied Sociolinguistics*. Cambridge, U.K.: Cambridge University Press, 1991, pp. 1–68.
- [15] B. Gambäck and A. Das, “Comparing the level of code-switching in corpora,” in *Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC’16)*, Portorož, Slovenia, 2016, pp. 1850–1855.
- [16] J. Li, M. Galley, C. Brockett, J. Gao, and B. Dolan, “A diversity-promoting objective function for neural conversation models,” in *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, 2016, pp. 110–119.
- [17] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, “BLEU: A method for automatic evaluation of machine translation,” in *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2002, pp. 311–318.
- [18] C.-Y. Lin, “ROUGE: A package for automatic evaluation of summaries,” in *Text Summarization Branches Out*, 2004, pp. 74–81.
- [19] M. Popović, “chrF: Character n-gram F-score for automatic MT evaluation,” in *Proceedings of the Tenth Workshop on Statistical Machine Translation*, 2015, pp. 392–395.
- [20] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, “BERTScore: Evaluating text generation with BERT,” in *International Conference on Learning Representations (ICLR)*, 2020.
- [21] J. Lin, “Divergence measures based on the Shannon entropy,” *IEEE Transactions on Information Theory*, vol. 37, no. 1, pp. 145–151, 1991.

APPENDIX

The prompts are rendered from `codemixesc/prompts.py` by `scripts/export_prompts.py` for an example Hindi-dominant seeker (62% Hindi words, CMI 0.38); braces mark the inputs. The parts that differ from MultiAgentESC are the *Language register* block, the reply instruction, the language and cultural fit criterion, the Register Gate prompt and the pivot’s back-translation; all other text is the base paper’s.

A. Emotion agent (cause and intention agents alike)

```
### Instruction
You are a psychological counseling expert. You will be
provided with a dialogue context between an 'Assistant'
and a 'User'. Please infer the emotional state
expressed in the user's last utterance.

### Dialogue context
{dialogue context}
```

```

### Language register
The user writes Hindi-dominant Hinglish (about 62% Hindi words) in Roman (Latin) script (code-mixing index 0.38).
Many Indian help-seekers mix Hindi and English (Hinglish). Interpret common Hinglish distress expressions (for example "tension", "ghabrahat", "log kya kahenge", "dimaag kharab", "mann nahi lagta") in context, by what the user means, rather than translating them literally.

Your answer must include the following elements:
Emotion: the emotion user expressed in their last utterance
.
Reasoning: the reasoning behind your answer.

Your answer must follow this format:
Emotion: [emotion]
Reasoning: [reasoning]

```

B. Response generator

```

You will be provided with a dialogue context between an 'Assistant' and a 'User'. Psychologists have analyzed the conversation, inferring the emotional state expressed by the user in their last utterance, the specific event that led to the user's emotional state and user's intention aiming to address the event that lead to their emotional state.

### Dialogue context
{dialogue context}

### Emotional state
{emotion agent output}

### Event
{cause agent output}

### Intention
{intention agent output}

### Language register
The user writes Hindi-dominant Hinglish (about 62% Hindi words) in Roman (Latin) script (code-mixing index 0.38).

Please generate a response from the Assistant's perspective using the {strategy} strategy. Reply in the same script with a similar Hindi-English mix, the way the user would naturally type it in chat. Write Hindi words in Roman script (for example 'bahut', 'pareshan'), never in Devanagari.
The following are examples of this strategy, all presented in the format of <post [strategy] response>. The examples are written in English; use them only to understand the strategy, not as a model of the language.

### Examples
{retrieved examples}

Your answer must be fewer than 30 words and must follow this format:
Response: [strategy] [response]

```

C. Debate

```

### You will be provided with a dialogue context between an 'Assistant' and a 'User'. Psychologists have analyzed the conversation, inferring the emotional state expressed by the user in their last utterance, the specific event that led to the user's emotional state and user's intention aiming to address the event that lead to their emotional state.

### Dialogue context
{dialogue context}

```

```

### Emotional state
{emotion agent output}

### Event
{cause agent output}

### Intention
{intention agent output}

### Language register
The user writes Hindi-dominant Hinglish (about 62% Hindi words) in Roman (Latin) script (code-mixing index 0.38).

Based on the provided information and dialogue context, please select the most appropriate response from the following options and explain why. Besides how well a response fits the conversation, the user's state and the support strategy, consider its language and cultural fit: does the response match the user's language, script and level of Hindi-English mixing, and does it sound natural?

### Response
{candidate responses}

Your answer must include the following elements:
Response: the most appropriate response and the strategy used in this response.
Reasoning: the reasoning behind your answer.

Your answer must follow this format:
Response: [strategy] [response]
Reasoning: [reasoning]

```

D. Reflection (voting round)

```

### You will be provided with a dialogue context between an 'Assistant' and a 'User'. Psychologists have analyzed the conversation, inferring the emotional state expressed by the user in their last utterance, the specific event that led to the user's emotional state and user's intention aiming to address the event that lead to their emotional state.

### Dialogue context
{dialogue context}

### Emotional state
{emotion agent output}

### Event
{cause agent output}

### Intention
{intention agent output}

### Language register
The user writes Hindi-dominant Hinglish (about 62% Hindi words) in Roman (Latin) script (code-mixing index 0.38).

Based on the provided information and the context of the dialogue, a group discussion is taking place to determine which response is the most appropriate.

### Discussion content
{debate}

You should carefully analyze the various different viewpoints above, reflect on your own thoughts, and ultimately arrive at a convincing result. Your thought can be changed if you believe the viewpoints of others are more reasonable. When judging the responses, also consider their language and cultural fit: does the response match the user's language, script and level of Hindi-English mixing, and does it sound natural?

Your answer must include the following elements:
Response: the most appropriate response and the strategy used in this response.

```

Reasoning: the reasoning behind your answer.

Your answer must follow this format:

Response: [strategy] [response]

Reasoning: [reasoning]

E. Refiner

You will be provided with a dialogue context between an 'Assistant' and a 'User'.

Dialogue context
{dialogue context}

Language register
The user writes Hindi-dominant Hinglish (about 62% Hindi words) in Roman (Latin) script (code-mixing index 0.38).

The following is a responses generated by the therapist using {strategy} strategy, presented in the format of <[strategy] response>. Please analyze whether this response is consistent with the ongoing conversation, whether it aligns with the strategy, whether it effectively helps alleviate the user's emotional stress, and whether it fits the user's language and culture: does the response match the user's language, script and level of Hindi-English mixing, and does it sound natural?

Response
[{strategy}] {response}

If the responses meets the above requirements, please return it as is; if not, please modify the response and provide a more refined version. Refined version must less than 30 words and must match the user's language register.

Your answer must include the following elements:
Response: original response or refined response and the strategy used in this response.
Reasoning: the reasoning behind your answer.

Your answer must follow this format:
Response: [strategy] [origianl/refined response]
Reasoning: [reasoning]

F. Register Gate

You will be provided with a dialogue context between an 'Assistant' and a 'User'.

Dialogue context
{dialogue context}

Language register
The user writes Hindi-dominant Hinglish (about 62% Hindi words) in Roman (Latin) script (code-mixing index 0.38).

The following response of the Assistant (which uses the {strategy} strategy) does not match the user's language : it has about 5% Hindi words, while the user's messages have about 62% (too little Hindi).

Response
[{strategy}] {response}

Rewrite the response so that it matches the user's language register. Reply in the same script with a similar Hindi-English mix, the way the user would naturally type it in chat. Write Hindi words in Roman script (for example 'bahut', 'pareshan'), never in Devanagari. Keep its meaning and the {strategy} strategy, and make it sound natural. The rewritten response must be fewer than 30 words.

Your answer must follow this format:
Response: [strategy] [rewritten response]

G. Translate-pivot: back-translation

Translate the following reply of an emotional support

Assistant from English into the user's language. The user writes Hindi-dominant Hinglish (about 62% Hindi words) in Roman (Latin) script (code-mixing index 0.38). Reply in the same script with a similar Hindi-English mix, the way the user would naturally type it in chat. Write Hindi words in Roman script (for example 'bahut', 'pareshan'), never in Devanagari. Keep the meaning, the support strategy and the warm tone, and keep it under 30 words.

Reply
{English response}

Your answer must follow this format:
Response: [translated reply]